



Maxim Kurkin

AI Researcher · Ph.D. Candidate at Skoltech

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SUMMARY

AI researcher and Ph.D. candidate specializing in training, evaluation, and safety of multimodal models (vision, language, audio). Track record of delivering research to production—from consulting the GigaChat team to co-founding an agentic AI startup. Built scalable benchmarks, compressed LLMs via novel Fisher approximations, and designed autonomous research pipelines. Passionate about teaching advanced deep learning and mentoring the next generation of researchers.

EXPERIENCE

Research Scientist

Jul 2023 – Present

[Artificial Intelligence Research Institute \(AIRI\), Moscow](#)

- Designed an infinitely scalable benchmark (MMReD) for dense-context LLM reasoning, questioning current test-time scaling paradigms
- Researched LLM compression via Kronecker-Factored Fisher Approximation for efficient deployment
- Developed a training framework for multimodal audiovisual LLM and OmniDialog benchmark for tri-modal understanding
- Enhanced OmniFusion MLLM with novel architectural components and curated synthetic datasets
- Consulting the GigaChat production team on integrating MLLM research

Research Engineer (concurrent)

Apr 2026 – Present

[Skoltech — NTI Center for Wireless Communications & IoT, Moscow](#)

- Contributing to the design of the 6G wireless communication standard

Head of AI, Co-Founder

Mar 2024 – Dec 2024

[ScreenMate AI, London](#)

- Defined technical vision and research roadmap for web QA and enterprise automation
- Managed a team of researchers and engineers across the full project lifecycle
- Architected the core agentic framework integrating VLMs with web interaction tools

Research Scientist

Apr 2022 – Jul 2023

[Bloomtech / UBIC Technologies, Moscow](#)

- Developed privacy-preserving ML algorithms using multi-party computation (MPC)
- Implemented secure distributed learning and inference for tabular, image, and speech data
- Led skull-to-face image matching project with domain adaptation and metric learning

MLE Intern

Jun 2021 – Sep 2021

[OZON.ru, Moscow](#)

- Implemented feature engineering logic for a large-scale product recommender system
- Automated CatBoost learning pipeline with MLFlow

PUBLICATIONS

- *NoReGeo: Non-Reasoning Geometry Benchmark*. AAAI 2025 Main Track. [Paper](#), [Code](#).
- *MMReD: A Cross-Modal Benchmark for Dense Context Reasoning*. ICLR 2026. [Paper](#), [Code](#), [Dataset](#). Press: [TASS](#), [Kommersant](#).
- *Fast and Accurate Fisher-Guided Quantization via Efficient Kronecker Factor Approximation*. Submitted @ ICLR 2026. [Preprint](#).
- *Generalized Fisher-Weighted SVD: Scalable Kronecker-Factored Fisher Approximation for Compressing LLMs*. Submitted @ ICLR 2026. [Preprint](#), [Code](#).
- *Binary Black-Box Optimization with Decision-Pretrained Transformer*. ICLR 2026 [SCI-FM Workshop](#). [Code](#).
- *OmniDialog: A Benchmark for Trimodal Dialogue Comprehension and Generation*. GenBench @ ACL 2024. [Paper](#).
- *OmniFusion Technical Report*. [Preprint](#).
- *A Method for Numerical Solution of a System of Stiff ODEs*. IITI 2021. [Paper](#).

TECHNICAL SKILLS

AI/ML LLM/MLLM Design Agentic AI Deep Learning ML System Design Convex Optimization

Tools PyTorch HuggingFace FSDP Triton vLLM Git Docker $\LaTeX$

Languages Russian (native) English (C1, TOEFL 106)

Research Spectral Methods Tensor Decomposition Fisher Information Bayesian Optimization
Privacy-Preserving ML

EDUCATION

Ph.D. in Computational and Data Science and Engineering Sep 2023 – Present

[Skolkovo Institute of Science and Technology \(Skoltech\)](#)

Advisor: Sergey Zagoruyko. Research on training, evaluation, and safety of multimodal models.

M.Sc. in Information Technology and Engineering Aug 2021 – Jun 2023

[Skoltech](#)

GPA: 4.9 / 5.0. Thesis on privacy-preserving federated learning; advised by Alexey Frolov.
Selected coursework: Deep Vision, NLP, Bayesian Methods, Cryptography.

B.Sc. in Applied Mathematics and Informatics Aug 2017 – Jun 2021

[Lomonosov Moscow State University \(MSU\)](#)

GPA: 4.6 / 5.0. Thesis on numerical optimization for ecological models.
Selected coursework: Functional Analysis, Numerical Methods, Mathematical Physics.

Additional Training

- **Ozon Masters / AI Masters**, Data Science (Sep 2020 – Jun 2023, GPA: 9.1 / 10.0)
- **AIRI Summer School** (Jul 2022) — Awarded **Best Project** for tensor decomposition optimization

TEACHING & MENTORSHIP

Instructor — Central University, Moscow — Advanced Deep Learning	Feb 2024 – Dec 2024
Instructor & Mentor — AIRI Summer School, Saint Petersburg	Aug 2024
Instructor — AI Masters (IAI MSU) — Deep Learning	Sep 2023 – Present
Guest Lecturer / TA — Skoltech — Transformers in Computer Vision	Oct 2023 – Dec 2023
Teaching Assistant — HSE — Deep Learning for Audio	Oct 2023 – Dec 2023

OTHER RESEARCH

HCCI Internal Combustion Engine *Mathematical Modeling, Control Systems*

Designed and tested a Homogeneous Charge Compression Ignition engine prototype. Developed a real-time mathematical model of engine control.

Boosting Neural Rendering Convergence *NeRF, Meta-Learning*

[Accelerated](#) NeRF training convergence via meta-learning pretraining strategies.

Normalizing Flows for Anomaly Detection *VAE, Normalizing Flows*

[Empirical study](#) of VAE with Fourier features stacked with normalizing flows for out-of-distribution detection.

Privacy in Federated Learning *MPC, Reed-Solomon Codes*

Master's thesis. [Implemented](#) generalized Lagrange encoding for multi-objective federated learning with complex-valued Reed-Solomon codes.